12

SEDIMENTATION

12.1 Introduction

The sedimentation of colloidal particles is important both in technology and in the laboratory. Gravity settlers, thickeners, or clarifiers commonly remove particles from waste streams issuing from a variety of processes. These generally operate as continuous processes that split the feed into two product streams, one the clear fluid and the other a sludge. Successful design requires knowledge of the sedimentation velocity of the particles over the relevant range of volume fractions and the role of interparticle forces in determining the structure of the dense sludge. Centrifugation provides a means of enhancing the driving force for commercial-scale operations, as well as concentrating or analyzing dispersions in the laboratory.

Despite their longstanding use, much remains to be understood about the details of processes which convert dilute dispersions into dense sediments. The key issues appear to be

(i) the variation of the settling velocity with volume fraction and interparticle potential,
(ii) the role of forces transmitted by interparticle potentials, and
(iii) the formulation of macroscopic models to predict the evolution of volume fraction as a function of position and time.

As with other colloidal phenomena, the complexity arises from the importance of a variety of interparticle forces and the fact that many systems of interest tend to be flocculated.

In this chapter we address all three points, but only for stable dispersions

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of spheres. At infinite dilution, a small sphere with density differing from that of the surrounding liquid by $\Delta\rho$ moves at the Stokes velocity,

$$U_0 = \frac{2a^2 \Delta\rho\, g}{9\mu},$$

determined by the balance between the gravitational force and the viscous drag (§2.6.5). Inertial effects remain unimportant, provided the Reynolds number is less than unity, i.e.

$$\begin{aligned} Re &= \frac{\rho a U_0}{\mu} \\ &= \frac{2a^3 \rho\, \Delta\rho\, g}{9\mu^2} < 1. \end{aligned}$$

Hereafter, we assume this to be satisfied, which for $\Delta\rho/\rho \approx 1$ in water requires $a < 75\,\mu\text{m}$.

At finite concentrations, hydrodynamic interactions can either increase or decrease the sedimentation velocity relative to U_0, depending on the nature of the non-hydrodynamic interactions. For purely repulsive interaction potentials, settling is hindered. Attractions, however, induce aggregation, causing two or more particles to settle as an effectively larger entity and, thereby, increasing the velocity. In both cases the net effect reflects the spatial distribution of particles, or the microstructure of the suspension, which depends on the range and the magnitude of the potential.

In the absence of an external field, the balance between the interparticle potential and Brownian motion completely determines the spatial distribution of particles in the suspension. But any relative velocity induced by an external field perturbs this microstructure. For example, for two particles with Stokes settling velocities differing by ΔU_0, the perturbation is characterized by a Peclet number for sedimentation,

$$\begin{aligned} Pe &= \frac{2a\, \Delta U_0}{D_0} \\ &= \frac{8\pi a^4 \Delta\rho g}{3kT} \frac{\Delta U_0}{U_0}. \end{aligned}$$

For aqueous systems with $\Delta\rho/\rho \approx 1$ and $\Delta U_0/U_0 \approx 1$, $Pe \approx 1$ for $1\,\mu\text{m}$ diameter particles. For $Pe \ll 1$, the perturbation is small, but for $Pe \gg 1$ the hydrodynamic forces dominate Brownian motion and determine the microstructure. Hence the concentration dependence of the sedimentation velocity for macroscopic particles can differ substantially from that for colloidal particles.

In a settling process, particles eventually accumulate at the bottom of the vessel, forming the sediment. Clearly, the boundary must exert a force to balance the weight of the particles. The transmission of this force upward through the dispersion represents the macroscopic consequence of thermodynamic interactions among particles in a non-uniform environment. The force acting on an individual particle is proportional to the gradient in its chemical potential, and, consequently, the gradient in the volume fraction. When balanced against gravity, this thermodynamic force determines the volume fraction profile within the sediment.

The following sections first address the dependence of the sedimentation velocity on the volume fraction and interparticle potential for both monodisperse and multimodal size distributions. Section 12.2 contains the formulation of the ensemble averages required to predict the average velocity from knowledge of interactions in dilute dispersions. The results are then described and compared with experimental data in §§12.3 and 12.4. The final two sections confront the questions of the thermodynamic force and macroscopic description of batch settling, for both incompressible sediments and stable but compressible ones.

12.2 Ensemble average velocities

The sedimentation velocity of particles in a concentrated suspension represents an average over all configurations of all particles. A prediction a priori, therefore, requires three elements: the velocities of particles for each configuration, the probability of observing an individual configuration, and a valid means of averaging. Since exact treatments of the hydrodynamic interactions exist only for two spheres, we restrict our attention to dilute concentrations for which pair interactions dominate. Then the pair distribution function characterizing the configurational probability can be determined from (8.3.8). Ensemble averaging (e.g., §3.2) is required to obtain the desired bulk property, but must be modified to accommodate the long-range hydrodynamic interactions.

Consider a representative volume V containing N equal spheres enclosed within a larger container of characteristic dimension L. The velocity at a point $\mathbf{x}$ within the volume is $\mathbf{U}$ if $\mathbf{x}$ resides within a particle and $\mathbf{u}$ otherwise. Both the particle and fluid velocities depend on the configuration of all N spheres. The respective ensemble averages, $\langle \mathbf{U} \rangle$ and $\langle \mathbf{u} \rangle$, will be in-

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dependent of the size and geometry of the container only if $L^3 \gg V \gg a^3/\phi$ so that $N \gg 1$.

At dilute concentrations, it is tempting to construct a virial expansion valid to $O(\phi)$ by expressing the velocity of the ith sphere through a pairwise additive approximation based on (2.8.1),

$$\begin{aligned}\mathbf{U}_i &= \mathbf{U}_0 + \sum_{\substack{j=1\\ \neq i}}^{N} \left[\boldsymbol{\omega}_{11}(\mathbf{r}_{ij}) - \frac{\boldsymbol{\delta}}{6\pi\mu a} + \boldsymbol{\omega}_{12}(\mathbf{r}_{ij}) \right] \cdot \mathbf{F}_j \\ &\equiv \mathbf{U}_0 + \sum_{\substack{j=1\\ \neq i}}^{N} \mathbf{U}_{ij}(\mathbf{r}_{ij}),\end{aligned} \tag{12.2.1}$$

with $\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j$. Ensemble averaging then yields

$$\begin{aligned}\langle \mathbf{U} \rangle &= \frac{1}{N!} \int P_N \mathbf{U}_1 \, d\mathbf{x}_1 \ldots d\mathbf{x}_N \\ &= \mathbf{U}_0 + n \int \mathbf{U}_{12}(\mathbf{r}) p(\mathbf{r}) \, d\mathbf{r} + O(\phi^2).\end{aligned} \tag{12.2.2}$$

Unfortunately $\boldsymbol{\omega}_{12}$ decays as $1/r$, while the pair distribution function $P_2(\mathbf{r}) = n^2 p(\mathbf{r})$ asymptotes to a constant, resulting in a divergent integral. Batchelor (1972) first resolved the problem and calculated the $O(\phi)$-correction to the Stokes velocity for hard spheres. Here we follow the approach of O'Brien (1979) as adapted by Glendinning & Russel (1982), one of several that have confirmed Batchelor's result.

The velocity at a point $\mathbf{x}$ in the fluid within the representative volume, expressed from (2.6.1) in terms of integrals over the surfaces of the particles $A_i (i = 1, \ldots, N)$ and a bounding surface S (Fig. 12.1) as described by O'Brien (1979), is given by

$$\mathbf{u}(\mathbf{x}) = -\int_{S+\Sigma A_i} \{\mathbf{I} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} - \mathbf{n} \cdot (\mathbf{u} \cdot \mathbf{J})\} \, d\mathbf{x}'. \tag{12.2.3}$$

$\mathbf{I}$ is the fundamental solution to the Stokes equations given by (2.5.16) for a point force and $\mathbf{J} = -3\mathbf{xxx}/4\pi x^5$ is the corresponding solution for a force dipole as deduced from (2.5.20); $\mathbf{n}$ is the unit normal directed into the fluid. The integrals over the A_i account for interactions among particles within the representative volume, while the integral over S represents the effect of particles outside. Strictly speaking, S should deivate from the surface enclosing V in order to bypass particles and remain in the fluid. However, in the dilute limit taking S to coincide with the surface of V introduces an error of only $O(\phi^2)$. Note that (12.2.3) comprises only a formal solution, since $\mathbf{u}$ depends on the unknown velocities and stresses on all the surfaces.

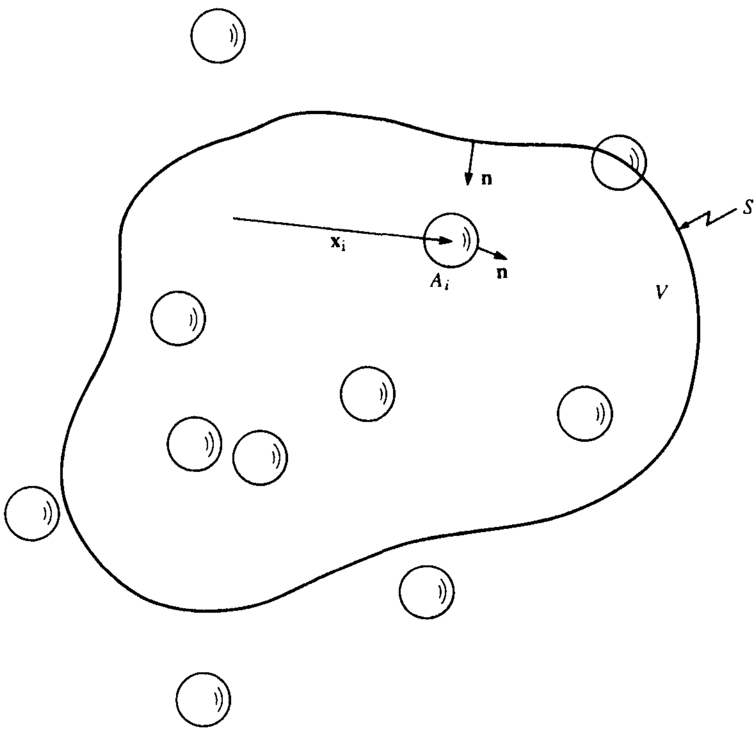
The velocity of the ith sphere follows from the local fluid velocity as

$$\mathbf{U}_i = \frac{1}{4\pi a^2}\int_{A_i} \mathbf{u}\,d\mathbf{x}. \tag{12.2.4}$$

Expanding $\mathbf{I}$ and $\mathbf{J}$ in Taylor series about the center of the ith sphere and employing the dilute approximation (12.2.1) for interactions with spheres within the representative volume yields

$$\mathbf{U}_i = \mathbf{U}_0 + \sum_{\substack{j=1 \\ \neq i}}^{N} \mathbf{U}_{ij} - \int_S \{\mathbf{I}\cdot\boldsymbol{\sigma}\cdot\mathbf{n} - \mathbf{n}\cdot(\mathbf{u}\cdot\mathbf{J}) + \frac{a^2}{6}\nabla^2\mathbf{I}\cdot\boldsymbol{\sigma}\cdot\mathbf{n}\}\,d\mathbf{x}'. \tag{12.2.5}$$

Fig. 12.1. Schematic of a representative volume V with surface S containing N particles at positions $\mathbf{x}_i$ with surfaces A_i. The unit normal $\mathbf{n}$ points into the fluid from S and all the A_i.



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Similarly, the fluid velocity can be written as

$$\mathbf{u} = \sum_{j=1}^{N} \mathbf{u}_j - \int_S \{\mathbf{I} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} - \mathbf{n} \cdot (\mathbf{u} \cdot \mathbf{J})\} \, d\mathbf{x}',$$

with (12.2.6)

$$\mathbf{u}_j = 6\pi\mu a \mathbf{U}_0 \cdot \left(1 + \frac{a^2}{6}\nabla^2\right)\mathbf{I}$$

representing the disturbance velocity generated by an isolated sphere.

For particles settling in an otherwise quiescent suspension, the average velocity of the suspension is zero, so

$$\mathbf{0} = \phi\langle \mathbf{U} \rangle + (1-\phi)\langle \mathbf{u} \rangle,$$

with (12.2.7)

$$\langle \mathbf{u} \rangle = \frac{1}{(N-1)!} \int \mathbf{u} \, P^*_{N-1} \, d\mathbf{x}_2 \ldots d\mathbf{x}_N,$$

where P^*_{N-1} specifies the probability of a particular configuration of the $N-1$ spheres given that $\mathbf{x}_1$ lies in the fluid. Thus $\langle \mathbf{u} \rangle = -\phi\langle \mathbf{U} \rangle/(1-\phi)$ represents the fluid velocity necessary to compensate for the volume flux of particles. The quantity of interest, the sedimentation velocity, is the difference between the ensemble averages of the particle and suspension velocities. Subtracting (12.2.7) from (12.2.5) leaves

$$\langle \mathbf{U} \rangle = \mathbf{U}_0(1-\phi) + \phi(\langle \mathbf{U}_{12} \rangle - \langle \mathbf{u}_1 \rangle) - \frac{a^2}{6} \int_S \nabla^2 \mathbf{I} \cdot \langle \boldsymbol{\sigma} \rangle \cdot \mathbf{n} \, d\mathbf{x} + O(\phi^2). \tag{12.2.8}$$

The ensemble average has been commuted with the integral, since $\mathbf{I}$ and $\mathbf{n}$ do not vary with the configuration of the particles.

Next, the bulk stress $\langle \boldsymbol{\sigma} \rangle$ must be derived from the microscopic momentum equation, which can be written as

$$\mathbf{0} = \nabla \cdot \boldsymbol{\sigma} + \begin{cases} \dfrac{9\mu}{2a^2}\mathbf{U}_0 & \text{within particles} \\ \mathbf{0} & \text{in the fluid.} \end{cases} \tag{12.2.9}$$

Straightforward ensemble averaging produces the macroscopic analogue

$$0 = \nabla \cdot \langle \boldsymbol{\sigma} \rangle + \frac{9\mu}{2a^2}\phi \mathbf{U}_0 \tag{12.2.10}$$

and integration determines

$$\langle \boldsymbol{\sigma} \rangle = -\frac{9\mu}{2a^2}\,\phi \mathbf{U}_0 \cdot \mathbf{x}\,\boldsymbol{\delta}. \tag{12.2.11}$$

This bulk stress simply accounts for the additional hydrostatic pressure due to the presence of the particles.

Substitution into (12.2.8) and use of the divergence theorem to convert the surface integral into a volume integral yields the sedimentation velocity as

$$\langle \mathbf{U} \rangle = \mathbf{U}_0 - \phi \mathbf{U}_0 - 6\pi\mu a n \mathbf{U}_0 \cdot \int_{r>a} \left(1 + \frac{a^2}{6}\nabla^2\right)\mathbf{I}(1-p)\,\mathrm{d}\mathbf{r} + \tfrac{1}{2}\phi \mathbf{U}_0 \tag{12.2.12}$$

$$+ 6\pi\mu a n \mathbf{U}_0 \cdot \int_{r>2a} \left\{\boldsymbol{\omega}_{11} - \frac{\boldsymbol{\delta}}{6\pi\mu a} + \boldsymbol{\omega}_{12} - \left(1 + \frac{a^2}{6}\nabla^2\right)\mathbf{I}\right\} p\,\mathrm{d}\mathbf{r} + O(\phi^2).$$

The terms in (12.2.12) have clear physical significance. At infinite dilution only the first, the *Stokes velocity*, remains. The second and third terms comprise the effect of '*backflow*', the reverse flow of fluid necessary to compensate for the volumetric flux of particles plus the associated fluid. The fourth term arises from the additional gradient in the *hydrostatic pressure*. The final integral, containing the mobilities, accounts for the *near-field hydrodynamic interactions*.

This formal expression, and the results presented earlier for the hydrodynamic mobilities and the pair distribution function, enable exact calculations of the sedimentation velocity to $O(\phi)$ in monodisperse suspensions. Furthermore, the generalization to multimodal dispersions is available (§12.4), and the extension to higher concentrations is possible, though limited by incomplete knowledge of the hydrodynamic interactions (e.g. Beenakker & Mazur, 1983).

12.3 Monodisperse suspensions of spheres

For two spheres of equal sizes and densities, the relative velocity

$$\Delta \mathbf{U} = (\boldsymbol{\omega}_{22} - \boldsymbol{\omega}_{21} - \boldsymbol{\omega}_{11} + \boldsymbol{\omega}_{12}) \cdot 6\pi\mu a \mathbf{U}_0$$

is identically zero, since $\boldsymbol{\omega}_{11} = \boldsymbol{\omega}_{22}$ and $\boldsymbol{\omega}_{12} = \boldsymbol{\omega}_{21}$. Thus at dilute concentrations the structure remains at equilibrium with $p(\mathbf{r}) = g(r) = \exp$

$(-\Phi/kT)$. With the explicit forms (2.8.4) for the mobilities and the spherical symmetry of $p(\mathbf{r})$, the expression for the sedimentation velocity reduces to

$$\langle \mathbf{U} \rangle = \mathbf{U}_0\left(1 - \phi\left\{5 + 3\int_2^\infty (1-g)s\,\mathrm{d}s - \right.\right.$$
$$\left.\left. - \int_2^\infty [A_{11} + A_{12} + 2(B_{11} + B_{12}) - 3(1 + s^{-1})]gs^2\,\mathrm{d}s\right\} + O(\phi^2)\right) \tag{12.3.1}$$
$$\equiv \mathbf{U}_0\{1 + K_2\phi + O(\phi^2)\},$$

with $s = r/a$.

For hard spheres with $g = H(s-2)$, the result (Batchelor, 1972)

$$\frac{U}{U_0} = 1 - 6.55\phi + O(\phi^2) \tag{12.3.2}$$

includes an $O(\phi)$-correction that consists of a relatively large negative contribution from the backflow (-5.5ϕ), a small positive effect from the pressure gradient $(+0.5\phi)$, and the hindrance due to the near field hydrodynamics (-1.55ϕ).

Experimental confirmation of this prediction requires demonstrating the hard sphere nature of the particle, e.g. $A_2 = 4$ (§10.3), as well as measuring K_2. For fd bacteriophage DNA, a compact spherical macromolecule with a hydrodynamic radius of 31.8 nm, Newman *et al.* (1974) measured a second virial coefficient of 4.0 ± 2.0 for the osmotic pressure and an $O(\phi)$-coefficient of -6.7 ± 0.8 for sedimentation. For the silica spheres shown to behave thermodynamically as hard spheres in §10.5, the corresponding value for sedimentation is -6.6 ± 0.6 (Kops-Werkhoven *et al.*, 1982).

The forms of the individual contributions to (12.3.1) suggest the effect of interparticle potentials. Attraction increases the population of nearby particles, thereby reducing the retardation from backflow while enhancing the modest contribution from near-field hydrodynamic interactions. Since the former is larger, attractions decrease in magnitude the negative $O(\phi)$-coefficient, increasing the sedimentation velocity. Conversely, a repulsion of longer range than the hard-sphere diameter causes greater retardation via backflow while reducing the near-field hydrodynamic interactions, thereby slowing sedimentation (Batchelor, 1972; Reed & Anderson, 1980; Batchelor & Wen, 1982).

As discussed in §10.3, the adhesive excluded-shell potential provides a useful model for demonstrating the combined effects of electrostatic repulsion and attraction into a secondary minimum. The radial distribution function

$$g(s) = H(s - s_0) + \frac{\delta(s - s_0)}{6\tau}, \tag{12.3.3}$$

together with the far-field approximations (2.8.9) for the mobilities yield

$$\frac{U}{U_0} = 1 - \left(K_0 - \frac{K_1}{\tau}\right)\phi + O(\phi^2),$$

with (12.3.4)

$$K_0 = \frac{3}{2}s_0^2 - 1 + \frac{15}{4s_0} - \frac{9}{8s_0^3} - \frac{107}{20s_0^5} + O(s_0^{-7}),$$

$$K_1 = \frac{1}{2}s_0 - \frac{5}{8s_0^2} + \frac{16}{9s_0^4} + \frac{107}{24s_0^6} + O(s_0^{-7}).$$

Both K_0 and K_1 are positive and increase monotonically with the excluded-shell radius. The two parameters, s_0 and $1/\tau$, also affect the second virial coefficient according to (10.3.7). Indeed, plotting K_2 vs. A_2 (Fig. 12.2)

Fig. 12.2. The $O(\phi)$-coefficient in the sedimentation velocity, K_2, as a function of the second virial coefficient in the osmotic pressure, A_2: ——, predictions for the adhesive excluded shell potential from (12.3.4) and (10.3.7); ●, data for silica spheres with $a = 32.5 \pm 0.2$ nm in toluene from Table 12.1. (Jansen, de Kruif, & Vrij, 1986).

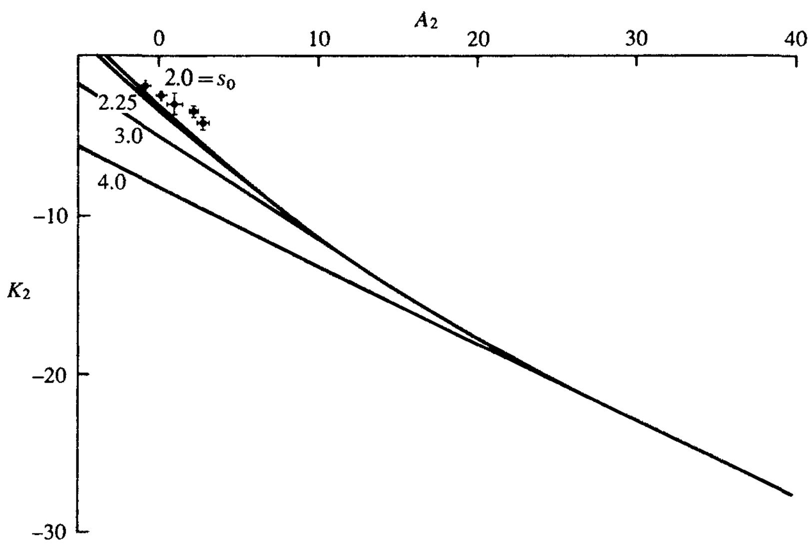

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Table 12.1. *Sedimentation of dilute suspensions with weak attractions*

(a) Silica spheres in toluene (Jansen, de Kruif & Vrij, 1986); $a = 32.5 \pm 0.2$ nm, $\Delta\rho = 890 \pm 10$ kg/m^3.

T/°C	A_2	K_2
25.0	2.86	-4.25 ± 0.39
20.0	2.19	-3.53 ± 0.31
14.8	1.02	-3.03 ± 0.68
12.1	0.11	-2.43 ± 0.15
10.0	-0.81	-1.89 ± 0.14

(b) Polystyrene latices in water

	Buscall *et al.* (1982)	Cheng & Schachman (1955)
a	1.55 μm	0.13 μm
I	10^{-3} M	10^{-1} M
ψ_0 (assumed)	-25 mV	-75 mV
$a\kappa$	155	130
s_0	2.04	2.04
$1/\tau$	0.23	2.22
A_2	4.0	1.9
K_2 (experiment)	-5.4 ± 0.1	-5.1
(theory)	-6.5	-5.0

produces a single curve for weak attractions ($1/\tau \leq 1.5$) or short-range repulsions ($s_0 - 2 \leq 0.25$). Under these conditions, which probably include most aqueous systems with significant secondary minima (Batchelor & Wen, 1982) and non-aqueous dispersions with interactions across thin polymer layers, the detailed form of the potential is unimportant.

Since the organophilic silica dispersions are stabilized by octadecyl chains grafted to their surfaces, weak attractions can be induced by exchanging the good solvent cyclohexane for a poor one such as toluene. Jansen, de Kruif, & Vrij (1986) measured the osmotic compressibility by light scattering and the sedimentation velocity with an analytical ultracentrifuge for particles of 32.5 nm radius in toluene over a range of temperatures. The values extracted for A_2 and K_2 indicate significant attractive interactions (Table 12.1) and the correlation between the two parameters closely resembles that predicted (Fig. 12.2).

Some data also exist for well-characterized aqueous systems. For example, Cheng & Schachman (1955) detected $K_2 = -5.06$ for 0.13 μm radius latices in 10^{-1} M NaCl and Buscall *et al.* (1982) found $K_2 = -5.4$ for

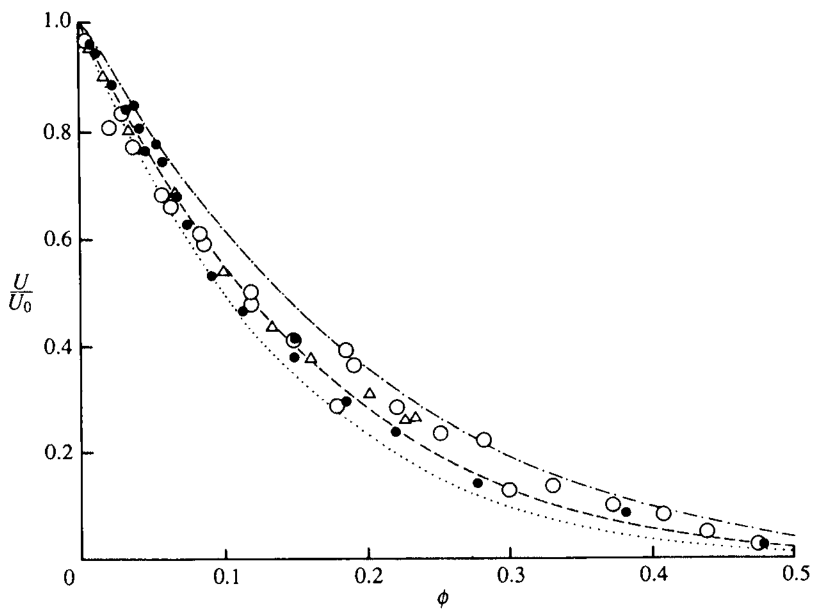
1.55 μm radius latices in 10^{-3} M NaCl. The values for s_0 and $1/\tau$ deduced with reasonable values for the unknown ψ_0 and the expected dispersion potential (§5.9, 5.10) generate the $O(\phi)$-coefficients listed in Table 12.1. In both cases A_2 and $-K_2$ lie at or below the values for hard spheres, indicating that the populations of doublets produced by the secondary minima outweigh the depletion of pairs at smaller separations due to the electrostatic repulsion.

At finite concentrations, the experimental results reported by Buscall *et al.* (1982) for the aqueous latices and de Kruif, Jansen, & Vrij (1987) for silica spheres in cyclohexane reflect the concentration dependence in the absence of either strong attractions or long range repulsions (Fig. 12.3). More accurate correlations can be constructed, but the simple expression,

$$\frac{U}{U_0} = (1-\phi)^{-K_2}, \tag{12.3.5}$$

with the exponent conforming to the $O(\phi)$ coefficient in the dilute limit, characterizes the behavior at both the dilute and the concentrated ends of

Fig. 12.3. The dimensionless sedimentation velocity as a function of volume fraction: ●, data for polystyrene spheres with $a = 1.55\,\mu$m in 10^{-3} M NaCl solution (Buscall *et al.*, 1982); ○, ▽, data for silica spheres with $a = 71 \pm 2$ nm and $a = 34.6 \pm 0.5$ nm in cyclohexane (de Kruif, Jansen, & Vrij, 1987): ···, (12.3.5) with $K_2 = -6.6$; - - -, (12.3.5) with $K_2 = -5.4$; -·-, (12.3.5) with $K_2 = -4.7$.



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the range. Thus, K_2 equals -6.6 for the silica dispersions and -5.4 for the aqueous latices (the experimental values). Both dispersions settle much like hard spheres; for stronger attractions or long-range repulsions, the volume fraction dependence might be quite different.

We began this section by noting that for dilute monodisperse spheres, sedimentation does not perturb the equilibrium microstructure. For higher-volume fractions or non-uniform particles, however, non-equilibrium effects can enter. Either three-body interactions or slight variations in particle size and/or density would generate a non-zero $\Delta\mathbf{U}$, thereby perturbing the pair distribution and altering the sedimentation velocity. Indeed, the volume fraction dependence of the sedimentation velocity for macroscopic spheres, represented by the third curve for $K_2=-4.7$ in Fig. 12.3, differs from that for colloidal spheres in qualitative accord with the dilute theory for unequal spheres described in the following section.

12.4 Polydisperse suspensions of spheres

Spheres of unequal sizes or densities settle at different velocities, producing a relative velocity for an interacting pair with magnitude scaling on

$$\Delta\mathbf{U}_0=\mathbf{U}_{0j}-\mathbf{U}_{0i}=(\gamma\lambda^2-1)\mathbf{U}_0$$

with $\gamma=\Delta\rho_j/\Delta\rho_i$, $\lambda=a_j/a_i$, and $U_0=U_{0i}$. Consequently, the pair distribution function will depart from equilibrium even in dilute suspensions. Generalization of the result of the previous section yields the average velocity for spheres of radius a_i, density difference $\Delta\rho_i$, and volume fraction ϕ_i interacting with spheres of $m-1$ other radii or densities as (Batchelor & Wen, 1982)

$$\langle\mathbf{U}_i\rangle-\langle\mathbf{u}\rangle=\mathbf{U}_{0i}\left(1+\sum_{j=1}^{m}K_{ij}\phi_j+O(\phi)\right),$$

with (12.4.1)

$$K_{ij}=-\gamma(\lambda^2+3\lambda+1)+\left(\frac{1+\lambda}{2\lambda}\right)^3\int_2^\infty\left\{(A_{11}+2B_{11}-3)p_{ij}\right.$$
$$\left.+2\gamma\frac{\lambda^3}{1+\lambda}\left[(A_{12}+2B_{12})p_{ij}-\frac{3}{s}\right]\right\}s^2\,ds,$$

where n^2p_{ij} is the distribution function for the ij pair and $s=2r/(a_i+a_j)$. The mobilities for unequal spheres are discussed in §2.8.

The pair conservation equation governing p_{ij}, a generalization of (8.3.8), accounts for relative motion due to the gravitational force, the interparticle potential, and Brownian motion. The variation of the coefficients with separation renders the equation difficult to solve analytically in any general sense, but semi-analytical solutions are possible for small and large values of the Peclet number (Batchelor & Wen, 1982). In the former case, a regular perturbation expansion suffices to determine the small departure from equilibrium. In the latter, Brownian motion is negligible, reducing the conservation equation to a first-order partial-differential equation that can be integrated along trajectories. The results define the asymptotic behavior of the sedimentation velocity for small and large particles. In this section we restrict our attention to the high Peclet number limit to demonstrate the effect of particle size on the behavior of nearly monodisperse suspensions and some qualitative features of bimodal systems.

For sufficiently large Peclet numbers, the interaction potential as well as Brownian motion can be neglected, reducing the steady-state conservation equation to

$$\nabla \cdot p_{ij}\mathbf{U}_{ij} = 0,$$

with

$$\mathbf{U}_{ij} = (\gamma\lambda^2 - 1)\mathbf{U}_0 \cdot \left[L\frac{\mathbf{rr}}{r^2} + M\left(\boldsymbol{\delta} - \frac{\mathbf{rr}}{r^2}\right)\right] \tag{12.4.2}$$

$$L = A_{ii} - \frac{2\lambda}{1+\lambda}A_{ij}, \quad M = B_{ii} - \frac{2\lambda}{1+\lambda}B_{ij},$$

from (2.8.2). The boundary condition

$$p_{ij} \to 1 \qquad \text{as} \qquad r_{ij} \to \infty$$

reflects the absence of long range structure.

For trajectories which start and end at infinite separation, the pair conservation equation can be integrated to determine (Batchelor, 1982)

$$p_{ij} = \exp \int_s^\infty \left(\frac{2(L-M)}{sL} + \frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}s} \right) \mathrm{d}s \tag{12.4.3}$$

Since p_{ij} depends only on the scalar separation, the hydrodynamic interactions preserve spherical symmetry but cause pairs to accumulate near contact. For equal spheres at infinite Peclet number, (12.4.3) yields different forms for p_{ij} at $\gamma = \lambda = 1$ according to how this limit is approached. The non-equilibrium distribution functions for both $\lambda = 1$, with $\gamma \to 1$, and $\gamma = 1$, with $\lambda \to 1$ (Fig. 12.4), resemble the equilibrium pair distribution resulting from a long range attractive potential, rather than $p_{ij} = H(s-2)$, which is characteristic of perfectly monodisperse hard spheres. Closed trajectories, which appear for small ranges of λ when the smaller sphere is more dense than the larger one, are not considered here.

For sufficiently large particles, even small variations in either density or radius lead to either

$$Pe = \frac{2aU_0}{D_0}(\lambda - 1) \gg 1 \qquad \text{for } \gamma = 1$$

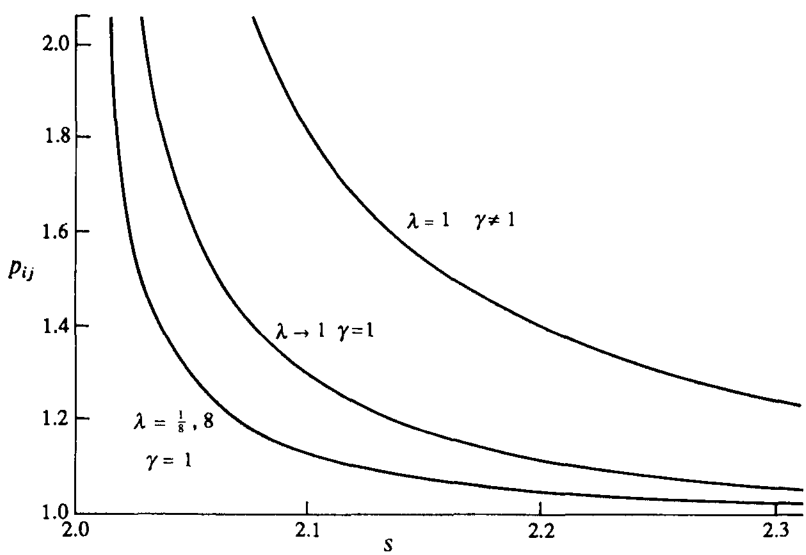
or

$$Pe = \frac{aU_0}{D_0}(\gamma - 1) \gg 1 \qquad \text{for } \lambda = 1.$$

For example, for glass beads of $10^2\,\mu$m radius in water, a 1 per cent variation in radius would produce a Peclet number of 10^7. Thus the volume-fraction dependence of the sedimentation velocity of large particles will not conform to that for a monodisperse suspension.

Substitution of (12.4.3) for the pair distribution into (12.4.1) determines the K_{ij}. The contributions can be separated into the far-field effects of backflow and the mean pressure gradient, the viscous drag due to a passive

Fig. 12.4. The dimensionless pair distribution function for $Pe = \infty$ for unequal spheres at several ratios of radii, λ, and density differences, γ, (Batchelor & Wen, 1982). Note the distinct difference between p_{ij} for $\lambda \to 1$ with $\gamma = 1$ and the value for identical spheres, $p_{ij} = 1$.



second sphere, and the velocity induced in the ith sphere by the motion of the jth. If the jth sphere is small, $\lambda \ll 1$, then only the viscous drag and backflow are significant, leaving

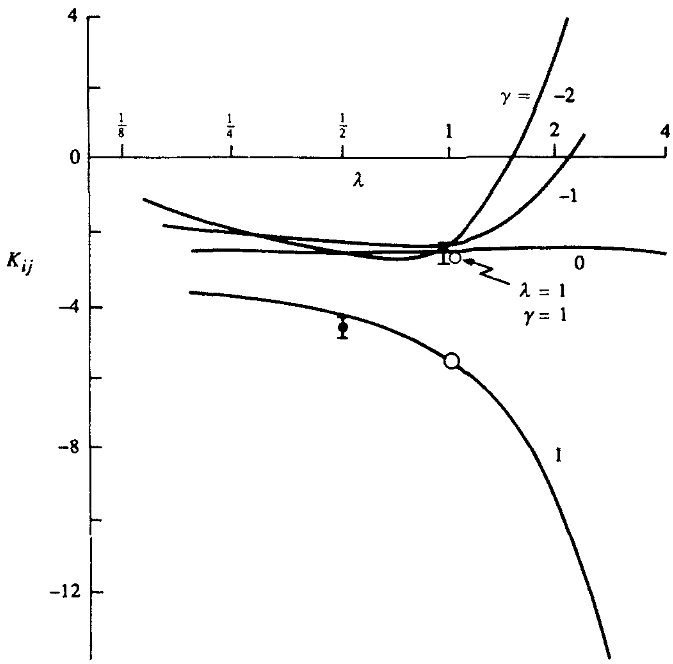
$$K_{ij} = -\tfrac{5}{2} - \gamma + O(\lambda). \tag{12.4.4}$$

Thus, the larger particle effectively settles through a continuum, with relative viscosity increased by 2.5ϕ and density altered by $\gamma\phi$. For the smaller sphere, only the far-field term contributes, giving

$$K_{ji} = -\frac{1}{\gamma}\left(1 + \frac{3}{\lambda} + \frac{1}{\lambda^2}\right) + O(\lambda). \tag{12.4.5}$$

So the smaller sphere is simply convected by the velocity and pressure field generated by the larger. Between these limits, the $O(\phi)$ coefficient varies with λ, as shown in Fig. 12.5. The open circle about $\lambda = \gamma = 1$ reflects the

Fig. 12.5. The coefficient characterizing the dependence of the sedimentation velocity of spheres of type i on the volume fraction of spheres of type j as a function of the size ratio $\lambda = a_j/a_i$ and the relative density difference $\gamma = \Delta\rho_j/\Delta\rho_i$ for $Pe = \infty$: ——, predictions of Batchelor & Wen (1982); ●, data of Davis & Birdsell (1988) for mixtures of glass beads ($\lambda = 0.52$, $\gamma = 1.0$) and glass and acrylic beads ($\lambda = 0.99$, $\gamma = 0.11$).



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singular nature of the point. The two data points from Davis & Birdsell (1988) for glass and acrylic spheres with $Pe \approx 10^7$ and $\gamma = 0.11$ with $\lambda = 1.0$ and $\gamma = 1.0$ with $\lambda = \frac{1}{2}$ agree with the predictions.

Despite the complicated form of Fig. 12.5, several aspects of the results are surprisingly simple and help to explain observations in the literature:

(i) For $\gamma = 0$, $K_{ij} = -2.5$ for all size ratios, indicating that neutrally buoyant spheres simply increase the effective viscosity of the medium. Recent experiments by Mondy, Graham, & Jensen (1986) find this prevails for heavy spheres settling in concentrated suspensions of neutrally buoyant spheres as well.

(ii) For equal sizes, K_{ij} depends weakly on the density ratio,

$$K_{ij} = -2.52 - 0.13\gamma, \tag{12.4.6}$$

except near $\gamma = 1$, where it jumps from -2.65 to -5.62.

(iii) When $\lambda \gg 1$ and $\gamma < 0$, $K_{ij} > 0$, indicating the backflow of the rising second species, which enhances the settling of the first, outweighs the near-field hydrodynamic interactions.

The different values for $\gamma \approx 1$ and $\lambda \approx 1$ provide some rationale for the difference between the settling velocities for large 'monodisperse' glass beads (Fig. 12.6), with

$$\frac{U}{U_0} = (1 - \phi)^{4.65}, \tag{12.4.7}$$

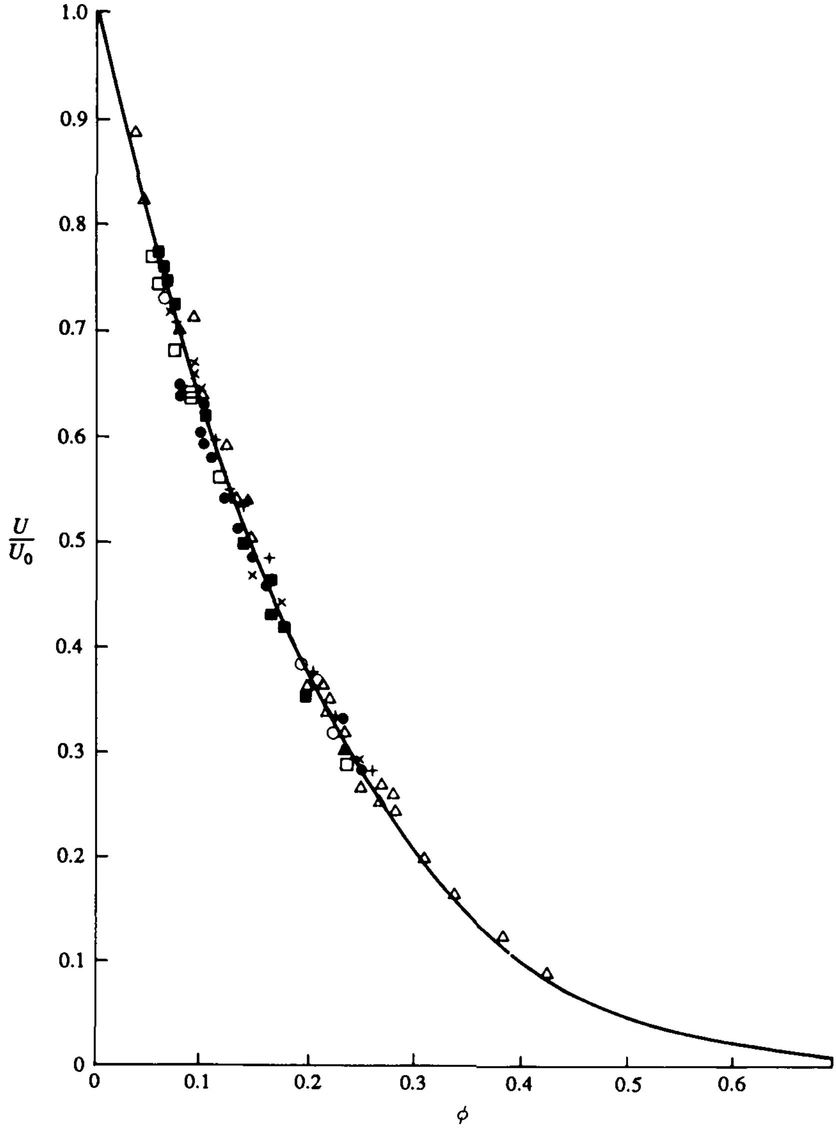
and the result noted earlier for monodisperse colloidal spheres (Fig. 12.3). For example, suppose one mixes equal parts of spheres with radii and density differences (1) a and $\Delta\rho$, (2) $a(1+\varepsilon)$ and $\Delta\rho$, and (3) a and $\Delta\rho(1+\varepsilon)$ to obtain a total volume fraction of ϕ. Then the average settling velocity would be

$$\begin{aligned}\frac{U}{U_0} &= \frac{U_1 + U_2 + U_3}{3U_0} \\ &= 1 + \tfrac{1}{9}\phi(K_{11} + K_{22} + K_{33} + 2K_{12} + 2K_{13} + 2K_{23}).\end{aligned} \tag{12.4.8}$$

With $K_{11} = K_{22} = K_{33} = -6.55$, $K_{12} = -5.62$, $K_{13} = -2.65$, and $K_{23} = -(5.62 + 2.65)/2 = -4.13$, this yields $K_1 = -4.92$, close to the value extracted from the data.

Though still limited, the theoretical and experimental results suggest a qualitative distinction between large and small particles and the effects of bimodal distributions. Some further information is available elsewhere (e.g. Davis & Acrivos, 1985). We now return our attention to monodisperse systems to examine the settling process.

Fig. 12.6. Dimensionless sedimentation velocity as a function of volume fraction for glass beads ($a = 0.35$ mm) in glycerine-water mixtures ($\mu = 0.001$–0.390 N s/m^2) such that $Re < 0.01$ and $aU_0/D_0 > 10^{11}$ (Hanratty & Bandukwala, 1957). The solid curve corresponds to (12.3.5), with $K_2 = -4.65$.



12.5 Theory of batch settling

As a dispersion settles in a closed container, the concentration necessarily becomes non-uniform (Fig. 12.7). A clear layer, devoid of particles, forms at the top and a sediment at the bottom. In addition, concentration gradients can appear in the region above the sediment. Kynch (1952) first formulated the theory capable of describing these phenomena for incompressible sediments. Here we generalize his treatment of monodisperse suspensions to account for compressibility imparted by the Brownian motion of and interactions among particles in the sediment.

In the absense of an external field, a spatially varying concentration represents a non-equilibrium state for a dispersion. Hence both particles and fluid molecules experience forces, proportional to the gradients in their respective chemical potentials (e.g. Batchelor, 1976),

$$\begin{aligned} \mathbf{F}_p &= -\nabla\mu_p, \\ \mathbf{F}_s &= -\nabla\mu_s. \end{aligned} \tag{12.5.1}$$

But a uniform body force, $-\mathbf{F}_s/l^3$ per unit volume, with l^3 the volume of a fluid molecule, acting on particles and fluid alike produces no relative motion between the two species. Hence the molecules can be treated as force-free if the effective force acting on each particle is written as

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_p - \frac{4\pi a^3}{3l^3}\mathbf{F}_s \\ &= -\nabla\left(\mu_p - \frac{4\pi a^3}{3l^3}\mu_s\right). \end{aligned} \tag{12.5.2}$$

Combining the Gibbs–Duhem relation,

$$n\nabla\mu_p + n_s\nabla\mu_s = 0, \tag{12.5.3}$$

with n and n_s the number densities of the particles and fluid molecules, respectively, and the volume-filling constraint,

$$\frac{4\pi a^3}{3}n + l^3 n_s = 1, \tag{12.5.4}$$

then yields

$$\begin{aligned} \mathbf{F} &= \frac{1}{nl^3}\nabla\mu_s \\ &= -\frac{1}{n}\nabla\Pi. \end{aligned} \tag{12.5.5}$$

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The final form for the thermodynamic force arising from the non-uniform concentration follows from the relationship (6.2.22) between Π, the osmotic pressure, and the chemical potential of the fluid.

With (12.5.5), the flux of particles due to the combined effects of gravity and the thermodynamic force can be written as

$$\phi\frac{K(\phi)}{6\pi\mu a}\left(\frac{4\pi a^3}{3}\Delta\rho\mathbf{g}+\mathbf{F}\right)=K(\phi)\left(\phi\mathbf{U}_0-D_0\frac{\mathrm{d}}{\mathrm{d}\phi}\,[\phi Z(\phi)]\nabla\phi\right),\tag{12.5.6}$$

with $Z(\phi)=\Pi/nkT$, the compressibility factor; $K(\phi)=U/U_0$, the sedimentation coefficient; and $D_0=kT/6\pi\mu a$, the Stokes–Einstein diffusion coefficient. Note that $K(\phi)/6\pi\mu a$ characterizes the mobility of particles in a concentrated dispersion. The conservation equation for a one-dimensional settling process then takes the form (Davis & Russel, 1988)

$$\frac{\partial\phi}{\partial t}+U_0\frac{\partial}{\partial x}\,\phi K(\phi)=D_0\frac{\partial}{\partial x}\left[K(\phi)\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi Z(\phi)]\frac{\partial\phi}{\partial x}\right],\tag{12.5.7}$$

with $\phi(x,0)=\phi_0$ and no flux boundary conditions at the top and bottom of the vessel, $x=0,h$.

Rendering the conservation equation dimensionless by scaling time on h/U_0 and the position on h identifies a single dimensionless group

$$Pe^{-1}=\frac{D_0}{hU_0}=\frac{3kT}{4\pi a^3\Delta\rho gh},$$

which multiplies the term on the right-hand side. This quantity, an inverse Peclet number for the macroscopic settling process, gauges the magnitude of the thermodynamic force relative to the gravitational force for a dilute dispersion. Settling in a conventional sense, i.e., to form a recognizable sediment, requires a large Peclet number. For example, for spheres of 1.0 μm radius with $\Delta\rho/\rho=1$ in a 1 m column of water the value is 10^7.

For large Peclet numbers the thermodynamic force is clearly negligible wherever $\phi Z(\phi)$ and the dimensionless gradient of the volume fraction are $O(1)$. Then, recasting the conservation equation as the quasi-linear first-order partial-differential equation

$$\frac{\partial\phi}{\partial t}+U_0\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi K(\phi)]\frac{\partial\phi}{\partial x}=0\tag{12.5.8}$$

leads to the general solution (cf. Rhee, Aris, & Amundson, 1987, §2)

$$\phi(x,t)=f(x-ut)$$

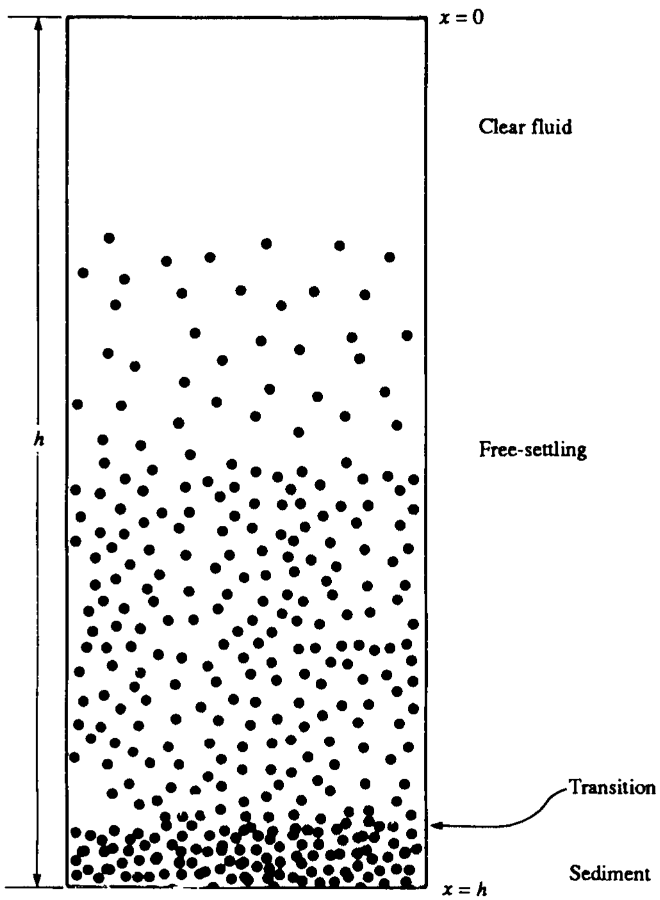
with (12.5.9)

$$u=U_0\frac{\mathrm{d}}{\mathrm{d}\phi}\phi K(\phi).$$

Consequently, in this free-settling region, ϕ is constant along characteristic curves with slopes $dx/dt = u$. The initial or boundary condition, where a characteristic intersects a coordinate axis, determines the value of ϕ.

The bottom of the container, however, comprises an impenetrable boundary at which the solution fails because the particle flux must vanish, even though $K(\phi)$ does not. Here particles accumulate until the volume fraction approaches closest packing. Since the compressibility factor diverges as $Z \sim (\phi_m - \phi)^{-1}$, as discussed in §10.5 (Woodcock, 1981), the thermodynamic force balances gravity only when $\phi_m - \phi = O(Pe^{-1})$. Then the time derivative becomes $O(Pe^{-1})$ smaller than the other terms. Invoking the pseudo-steady approximation, integrating the conservation

Fig. 12.7. Schematic for a batch settling process in a container of height h.



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equation once, and applying the no-flux boundary condition at the bottom simply leaves

$$\frac{4\pi a^3}{3}\mathbf{g} = \frac{1}{n}\nabla\Pi. \tag{12.5.10}$$

Thus the thermodynamic force supports particles. In the bulk of the sediment this occurs because the osmotic pressure is large. But at the top of the sediment, the volume fraction changes from near closest packing to the value in the free-settling region. This produces a thin transition layer in which the thermodynamic forces are significant because $\partial\phi/\partial x = O(Pe)$.

These arguments identify three regions within the settling dispersion (Fig. 12.7):

(i) free-settling $\quad \phi_m - \phi = O(1) \quad \dfrac{\mathrm{d}Z(\phi)}{\mathrm{d}\phi} = O(1)$

$$\frac{\partial\phi}{\partial x} = O(1);$$

(ii) transition $\quad \phi_m - \phi = O(1) \quad \dfrac{\mathrm{d}Z(\phi)}{\mathrm{d}\phi} = O(1)$

$$\frac{\partial\phi}{\partial x} = O(Pe).$$

(iii) sediment $\quad \phi_m - \phi = O(Pe^{-1}) \quad \dfrac{\mathrm{d}Z(\phi)}{\mathrm{d}\phi} = O(Pe^2)$

$$\frac{\partial\phi}{\partial x} = O(Pe^{-1})$$

In the limit of $Pe \to \infty$ the sediment becomes incompressible, i.e. $\phi \to \phi_m$, and the transition region shrinks to a discontinuity. In the following two sections, we address first the case of $Pe = \infty$ and then the expansion of the sediment due to the thermodynamic forces at finite Peclet numbers.

12.6 Hard spheres at infinite Peclet number

Letting $Pe \to \infty$ eliminates the flux due to the thermodynamic forces and reduces the order of the differential equation. Thus the limit is singular, since the solutions develop discontinuities and the governing equation ceases to be valid within the sediment. These difficulties are resolved by setting $\phi = \phi_m$ within the sediment and integrating the differential equation across discontinuities (Kynch, 1952).

For a discontinuity at $x=L(t)$, the coordinate transformation $y=(x-L(t))Pe$ converts the dimensionless version of (12.5.7) into

$$\frac{1}{Pe}\frac{\partial\phi}{\partial t}+\frac{\partial}{\partial y}\phi(K(\phi)-L')=\frac{\partial}{\partial y}K(\phi)\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi Z(\phi)]\frac{\partial\phi}{\partial y}. \qquad (12.6.1)$$

Integrating the pseudosteady $O(1)$ equation with $\phi\to\phi^+$ and $\partial\phi/\partial y\to 0$ as $y\to-\infty$ yields

$$\phi(K(\phi)-L')-\phi^+(K(\phi^+)-L')=K(\phi)\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi Z(\phi)] \qquad (12.6.2)$$

Evaluating (12.6.2) as $y\to\infty$ then determines $L'\equiv \mathrm{d}L/\mathrm{d}t$ as follows:

(i) $\phi\to\phi_m$, so that

$$\lim_{y\to\infty}K(\phi)\left[\phi-\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi Z(\phi)]\frac{\partial\phi}{\partial y}\right]=0,$$

leaving

$$L'=-\frac{\phi^+K(\phi^+)}{\phi_m-\phi^+} \qquad (12.6.3)$$

as the rate of rise of the sediment.

(ii) $\phi\to\phi^-$ and $\partial\phi/\partial y\to 0$, so that

$$L'=\frac{\phi^+K(\phi^+)-\phi^-K(\phi^-)}{\phi^+-\phi^-} \qquad (12.6.4)$$

for a discontinuity in the free-settling region.

The nature of permissible discontinuities follows from substituting (12.6.3) or (12.6.4) into (12.6.2). For the latter, this produces

$$\begin{aligned}K(\phi)\frac{\mathrm{d}}{\mathrm{d}\phi}[\phi Z(\phi)]\frac{\partial\phi}{\partial y}&=\phi K(\phi)-\frac{(\phi-\phi^-)\phi^+K(\phi^+)-(\phi-\phi^+)\phi^-K(\phi^-)}{\phi^+-\phi^-}\\&=\Delta\phi K(\phi).\end{aligned} \qquad (12.6.5)$$

The right-hand side is simply the vertical distance between points at a particular ϕ on the flux curve and the line connecting points above and below the discontinuity, i.e. $(\phi^+, K(\phi^+))$ and $(\phi^-, K(\phi^-))$ in Fig. 12.8. Since $\partial\phi/\partial y\to 0$ as $y\to\pm\infty$ but $\partial\phi/\partial y\neq 0$ for $-\infty<y<+\infty$, (12.6.5) proves that the line cannot cross the flux curve. Hence the discontinuity depicted in Fig. 12.8 is impossible; the line must lie entirely below (above) the flux curve for $\partial\phi\partial y>(<)0$, in accord with the entropy condition of Lax (e.g., Rhee, Aris, & Amundson, 1987, §5.4).

To demonstrate the features of the settling process, we consider disordered hard-sphere dispersions for which $\phi_m = 0.64$ and

$$K(\phi) = (1-\phi)^{6.55}, \qquad (12.6.6)$$

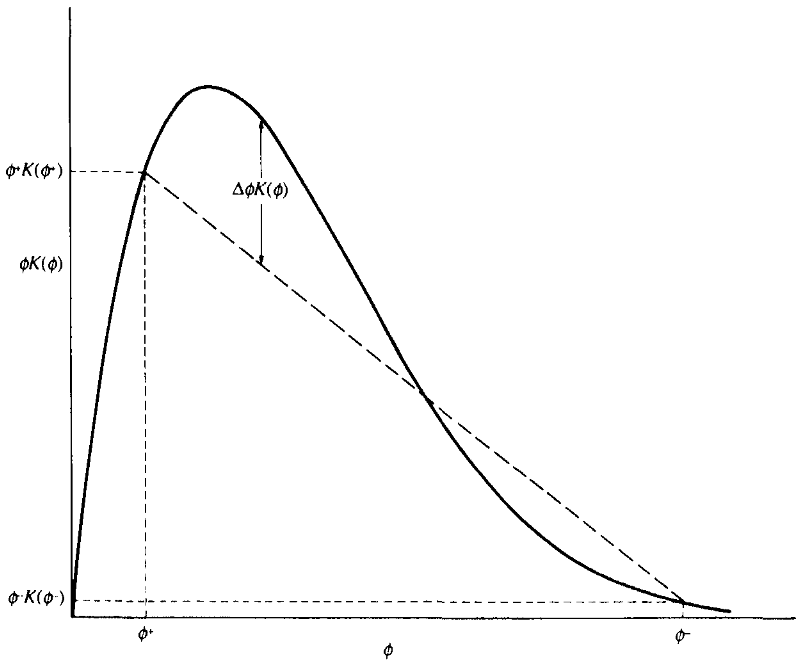
so the flux curve takes the form shown in Fig. 12.9. Tangents to this curve then determine the slope of the characteristics, $dx/dt = u$, and the velocity of the rising sediment corresponds to the slope of straight lines intersecting the horizontal axis at ϕ_m and the flux curve at ϕ_+. Two noteworthy features of the flux curve are:

(i) the inflection point at $\phi = 0.265$, separating Regions I and II, where $du/d\phi < 0$, from Regions III and IV, where $du/d\phi > 0$; and
(ii) the line from ϕ_m that is tangent to the flux curve at $\phi = 0.565$ and intersects the curve at $\phi = 0.024$.

Note that for $K(\phi)$ of the form (12.3.5), Regions II and III disappear for

$$-K_2 < \frac{2}{\phi_m}[1 + (1-\phi_m)^{1/2}] - 1,$$

Fig. 12.8 Discontinuity between regions settling freely with volume fractions ϕ^+ and ϕ^-.



since the point of tangency vanishes. The significance of these four regions of the flux curve will be explained below.

The simplest and most familiar case is for a uniform volume fraction in each of the three regions – clear fluid, freely settling suspension, and sediment – such that

$$\phi = \begin{cases} 0 & 0 < x < U_{\text{top}} t \\ \phi_0 & U_{\text{top}} t < x < U_{\text{sed}} t \\ \phi_m & U_{\text{sed}} t < x < h \end{cases} \tag{12.6.7}$$

The discontinuities between the clear fluid and the settling suspension and the settling suspension and the sediment move with velocities

$$U_{\text{top}} = U_0 \frac{\phi_0 K(\phi_0)}{\phi_0} = U_0 K(\phi_0) \tag{12.6.8}$$

from (12.6.4) and

$$U_{\text{sed}} = -U_0 \frac{\phi_0 K(\phi_0)}{\phi_m - \phi_0} \tag{12.6.9}$$

Fig. 12.9. Flux curve for hard spheres constructed from (12.3.5), with $K_2 = -6.55$ and $\phi_m = 0.64$. The four regions defined by the broken lines correspond to: I, $0 \le \phi \le 0.024$; II, $0.024 \le \phi \le 0.264$; III, $0.264 \le \phi \le 0.565$; IV, $0.565 \le \phi \le 0.64$.

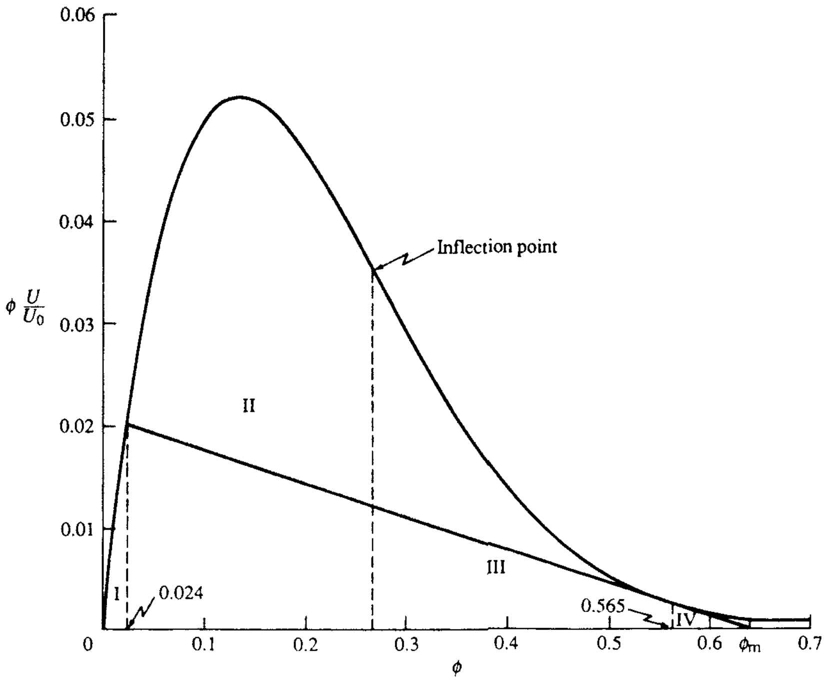


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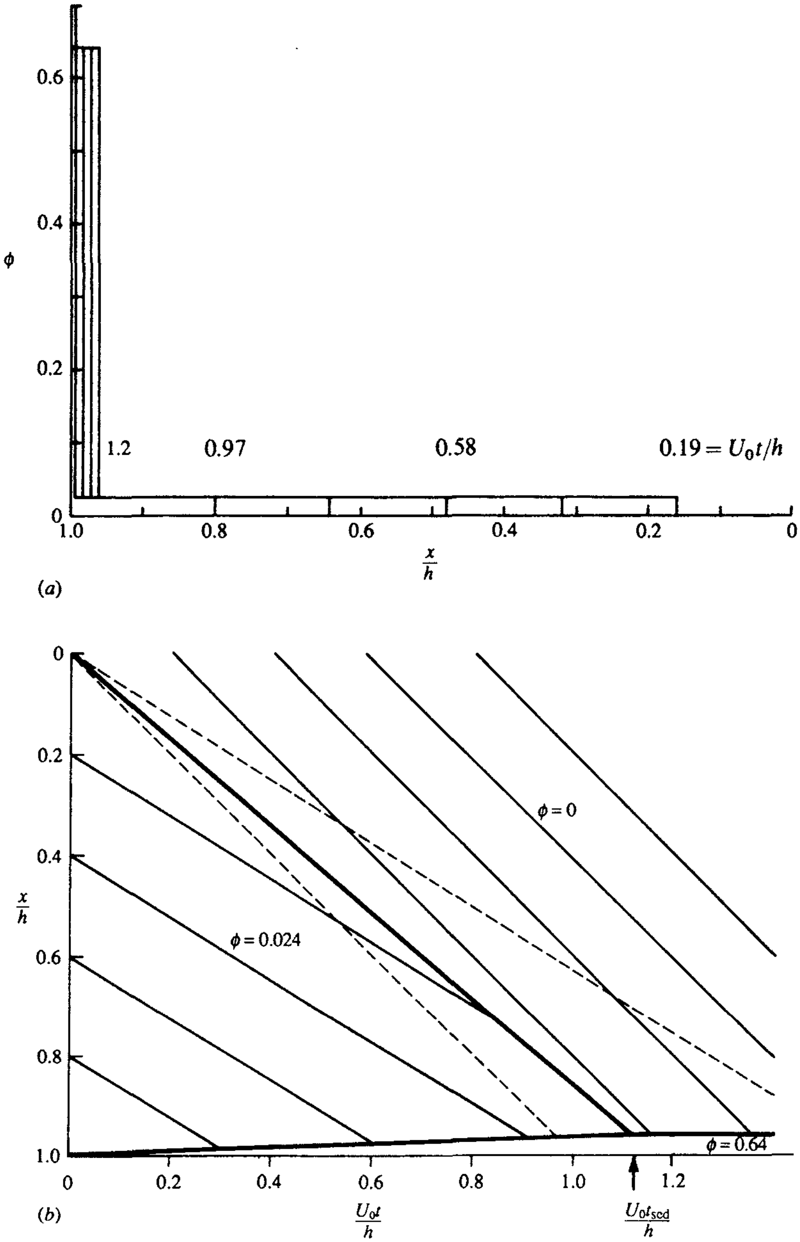
Fig. 12.10. Solutions for batch settling of hard spheres with $Pe = \infty$ and $\phi_0 = 0.024$: (*a*) Volume fraction as a function of position at several times. (*b*) Contour plot, showing: heavy unbroken line, discontinuities; light unbroken line, characteristics along which ϕ = const.; broken line, characteristics eliminated by discontinuities.



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from (12.6.3), with $\phi^+ = \phi_0$, respectively. Thus the volume fraction varies as shown in Fig. 12.10(a) for $\phi_0 = 0.024$ and the sediment reaches its final height, $h\phi_0/\phi_m$, at time $h(1-\phi_0/\phi_m)/U_0K(\phi_0)$.

The construction of this solution via the method of characteristics proceeds as follows. The first step is to plot the characteristic curves originating at the boundaries of the domain where the initial and boundary conditions set ϕ (Fig. 12.10(b)). If these characteristics fill the entire $x-t$ plane without intersecting then the solution is complete and single-valued. Generally, however, the characteristics intersect as shown, implying an unacceptable multiplicity of solutions. The second step then is to eliminate this multiplicity by constructing one or more discontinuities, separating regions in which the solution varies continuously and propagating at velocities determined by (12.6.2). The corresponding chord on Fig. 12.9 lies below the flux curve for $\phi_0 \leq 0.024$.

For values of ϕ_0 in Regions II and III, though, the chord connecting ϕ_0 on the flux curve with ϕ_m on the horizontal axis crosses the flux curve (Fig. 12.11(a)). So the jump in volume fraction from ϕ_0 to ϕ_m at the top of the sediment is inadmissable. Instead, solutions with discontinuities represented by chords lying below the flux curve must be constructed by recognizing that additional characteristics emanate from the origin with slopes corresponding to $\phi_0 \leq \phi \leq \phi_m$.

For example, in Region II a fan of characteristics from the origin (Fig. 12.11(b)) is bounded above by a discontinuity with the same velocity, i.e.

$$U_0 \frac{\phi_1 K(\phi_1) - \phi_0 K(\phi_0)}{\phi_1 - \phi_0} = U_0 \left. \frac{\mathrm{d}[\phi K(\phi)]}{\mathrm{d}\phi} \right|_{\phi=\phi_1}. \tag{12.6.10}$$

For $\phi_0 = 0.1$, this tangency condition determines $\phi_1 = 0.40$. The volume fraction at the bottom of the fan, $\phi^+ = 0.565$, is determined by the analogous condition

$$-U_0 \frac{\phi^+ K(\phi^+)}{\phi_m - \phi_+} = U_0 \left. \frac{\mathrm{d}[\phi K(\phi)]}{\mathrm{d}\phi} \right|_{\phi=\phi^+}. \tag{12.6.11}$$

Now the solution takes the form shown in Fig. 12.11(c), with a region of smoothly varying volume fraction determined by

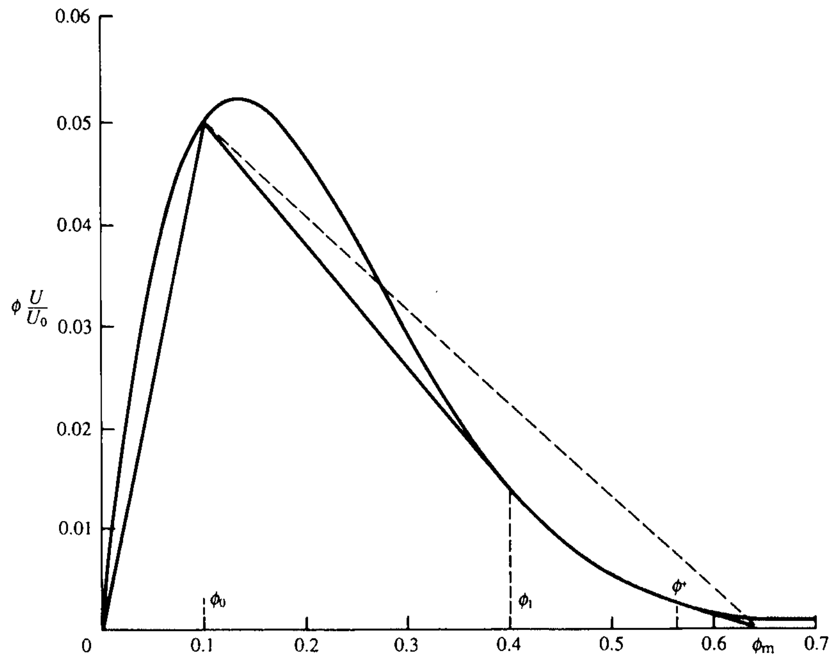
$$\frac{x}{t} = -\frac{\mathrm{d}[\phi K(\phi)]}{\mathrm{d}\phi} \tag{12.6.12}$$

with $\phi_1 \leq \phi \leq \phi^+$, separating the uniform region from the sediment. Since the volume fraction increases through the fan, U_{top} decreases in the final stage of the process, termed the falling-rate period. The behavior in Region

III is similar, except that the discontinuity at the top of the fan vanishes so that $\phi_1 = \phi_0$.

Table 12.2 summarizes the key features of these solutions for incompressible sediments: the volume fractions at the top (ϕ_1) and bottom (ϕ^+) of the fan, the beginning of the falling-rate period (t_1), and the time for complete sedimentation (t_{sed}). The solutions are unique, provided all chords representing discontinuities lie below the flux curve and no characteristics emerge from the upper surface of the sediment, which bounds the domain of the solution. The validity of these assumptions is proven by the asymptotic arguments above, and is supported by the numerical solutions for finite Peclet numbers described in the next section.

Fig. 12.11. Solutions for batch settling of hard spheres, with $Pe = \infty$ and $\phi_0 = 0.10$: (*a*) Flux curve with chords corresponding to the inadmissable discontinuity $\phi_0 \rightarrow \phi_m$ and the three discontinuities required for the correct solution: $0 \rightarrow \phi_0$ at the top of the free-settling region; $\phi_0 \rightarrow \phi_1$ at the top of the fan; $\phi^+ \rightarrow \phi_m$ at the top of the sediment. (*b*) Contour plot, showing: heavy line, discontinuities; lighter line; characteristics; the beginning of the falling-rate period at $U_0 t_1/h$, and the end of the process at $U_0 t_{sed}/h$. (*c*) Volume fraction as a function of position and time.



(*a*)

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12.7 Hard spheres at finite Peclet number

Direct numerical solutions of the full equation (12.5.7), with suitable forms for $K(\phi)$ and $Z(\phi)$, serve to illustrate the effects of the thermodynamic forces at finite Peclet number and confirm the arguments above as $Pe \to \infty$. Since large Peclet numbers are of primary interest, the thermodynamic forces become important only as $\phi \to \phi_m$. Although hard spheres at equilibrium exist in a face-centered-cubic solid phase for $\phi > 0.55$, computer simulations (Woodcock, 1981) and experiments (Pusey & van Megen, 1987) reveal a persistent disordered state up to random close packing at $\phi \approx 0.64$. So, the calculations described below pertain to a disordered fluid with (12.6.4) for $K(\phi)$ and

$$Z(\phi) = \frac{1.85}{\phi_m - \phi}, \tag{12.7.1}$$

with $\phi_m = 0.64$ in accord with the results of Woodcock (1981).

Results for $\phi_0 = 0.1$ (Auzerais, Jackson, & Russel, 1988) illustrate several points clearly:

(i) For $Pe = 850$, the thermodynamic forces smooth the discontinuities at the top of the uniform settling region only slightly but completely obliterate the distinction between the sediment and the fan (Fig. 12.12(*a*)). Indeed, the volume fraction within the sediment approaches ϕ_m only at long times, and then only at the very bottom.

(ii) Nonetheless, the contour plot of the solution (Fig. 12.12(*b*)) maintains the same qualitative form as at infinite Peclet number, with the top of the suspension, defined by the contour for $\phi = \phi_0/2 = 0.05$, falling at the velocity $U(\phi_0)$ independent of Pe.

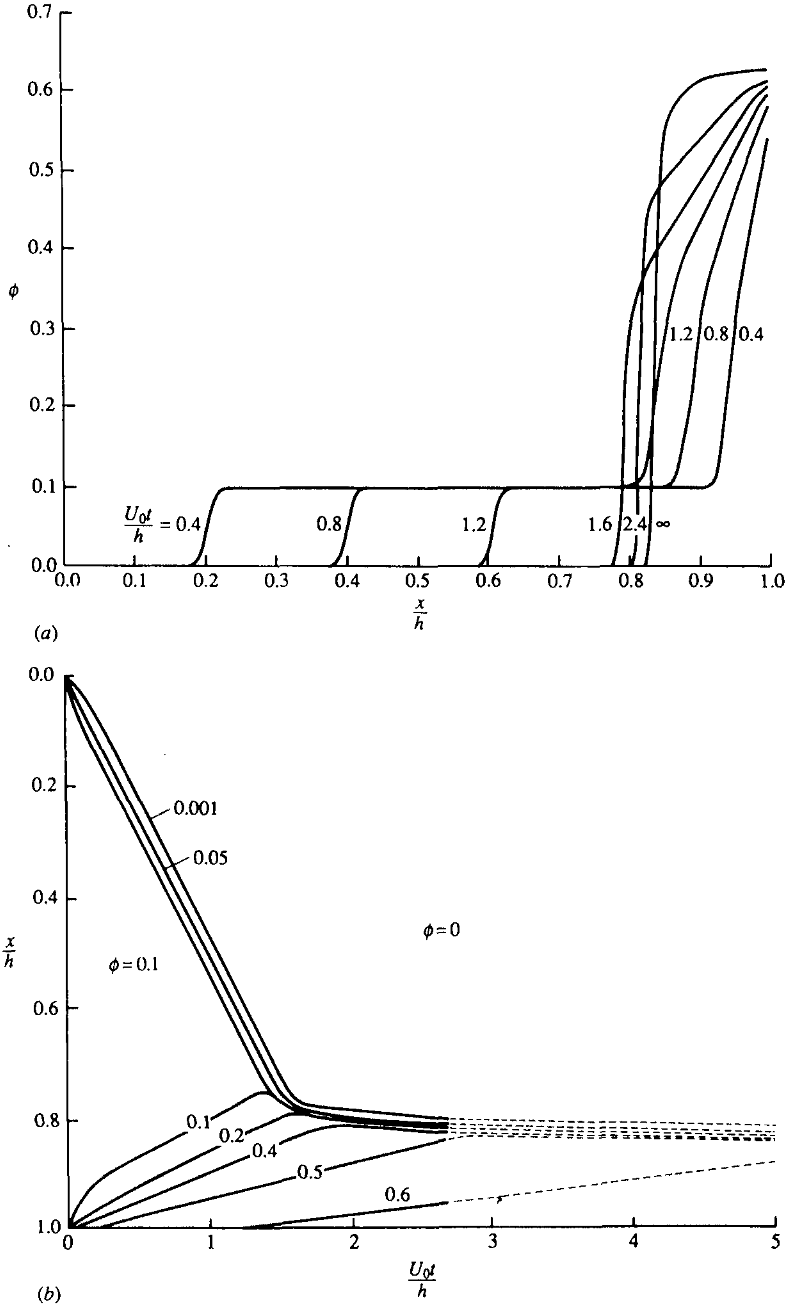
(iii) For $Pe = 175\,000$, the solution (Fig. 12.13) does approach the analytical solution for $Pe = \infty$, though the discontinuity at the top of the sediment remains somewhat expanded.

The gradual transition from the fan to the sediment, even for $Pe \approx 10^5$, reflects the difference between the thermodynamic force at ϕ^+ and that at infinite dilution. The actual ratio of thermodynamic to gravitational forces at the top of the sediment,

$$\frac{D_0}{hU_0} \frac{\mathrm{d}[\phi Z(\phi)]}{\mathrm{d}\phi}\bigg|_{\phi+} = \frac{210}{Pe},$$

explains the considerably thicker transition layer. The approach of the numerical solutions to the analytical solution as $Pe \to \infty$, together with the asymptotic arguments mentioned above, affirm the validity of the results of §12.6.

Fig. 12.12. Numerical solutions for batch settling of hard spheres, with $Pe = 850$ and $\phi_0 = 0.10$ (Auzerais, Jackson, & Russel, 1988): (*a*) Volume fraction as a function of position and time. (*b*) Contour plot.



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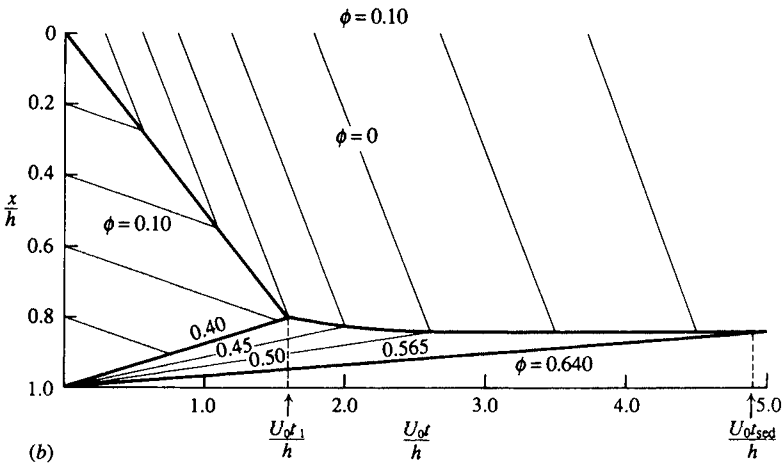
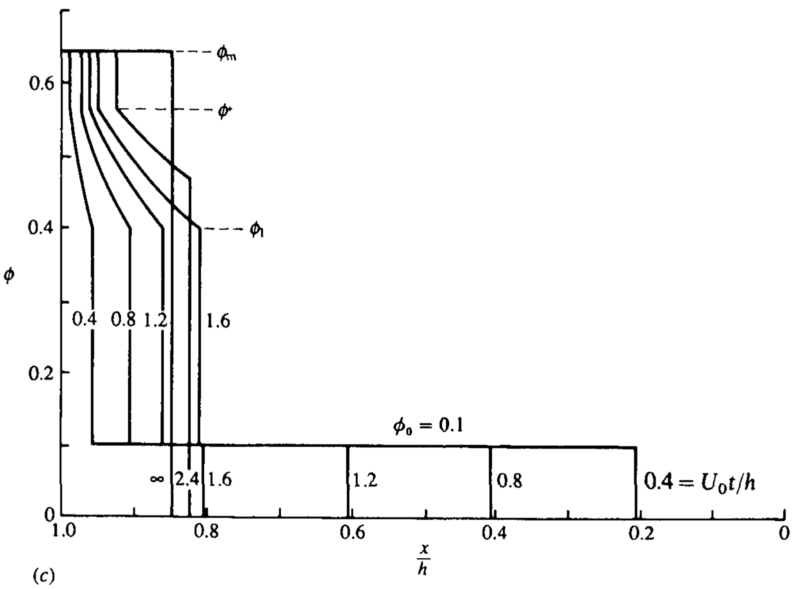


(b)



(c)

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Table 12.2. *Characteristics of batch settling at $Pe = \infty$*

Region	ϕ_1	ϕ^+	t_1	t_{sed}
I, IV	ϕ_0	ϕ_m	t_{sed}	$\dfrac{(1-\phi_0/\phi_m)}{K(\phi_0)}$
II	$\dfrac{\phi_1 K(\phi_1)-\phi_0 K(\phi_0)}{\phi_1-\phi_0}=\dfrac{d(\phi_1 K)}{d\phi_1}$	$-\dfrac{\phi^+ K(\phi^+)}{\phi_m-\phi^+}=\dfrac{d(\phi^+ K)}{d\phi^+}$	$\dfrac{(1-\phi_0/\phi_1)}{K(\phi_0)-K(\phi_1)}$	$\dfrac{\phi_0(1-\phi^+/\phi_m)}{\phi^+ K(\phi^+)}$
III	ϕ_0	$-\dfrac{\phi^+ K(\phi^+)}{\phi_m-\phi^+}=\dfrac{d(\phi^+ K)}{d\phi^+}$	$\dfrac{1}{K(\phi_0)-\dfrac{d(\phi_0 K)}{d\phi_0}}$	$\dfrac{\phi_0(1-\phi^+/\phi_m)}{\phi^+ K(\phi^+)}$

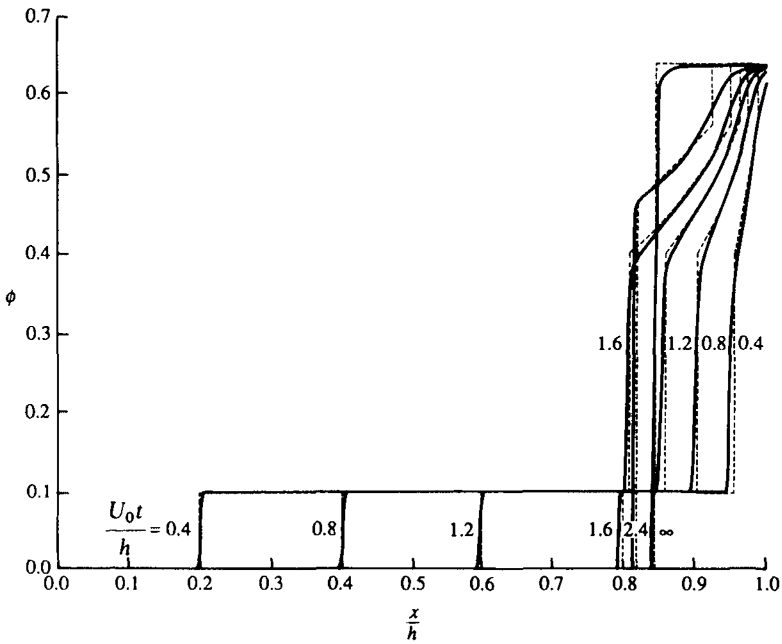
12.8 Summary

This chapter has highlighted the effects of interactions, both hydrodynamic and thermodynamic, on the sedimentation velocity and the transient settling process. The pair interaction theory for the velocity is exact at dilute concentrations and provides a basis for understanding and correlating the behavior of concentrated systems. In particular, it identifies the effects of attractive and repulsive potentials and the non-equilibrium structure responsible for the distinction between the velocities of colloidal and macroscopic particles. The treatment of batch settling demonstrates the classical theory for incompressible sediments to be the $Pe \rightarrow \infty$ limit for stable dispersions.

The basic approach has the potential for addressing several other interesting and important problems:

(i) the volume fraction dependence of the velocity in concentrated dispersions with stronger interparticle potentials or multimodal size distributions,

Fig. 12.13. Volume fraction as a function of position and time from numerical solutions for batch settling of hard spheres, with $Pe = 175000$ and $\phi_0 = 0.10$ (Auzerais, Jackson, & Russel, 1988).



(ii) the behavior of non-spherical particles,
(iii) the settling of flocculated or phase separated systems (Buscall & White, 1987).

Further progress requires descriptions of the structure and the hydrodynamic interactions under more complex conditions than considered here.

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Problems

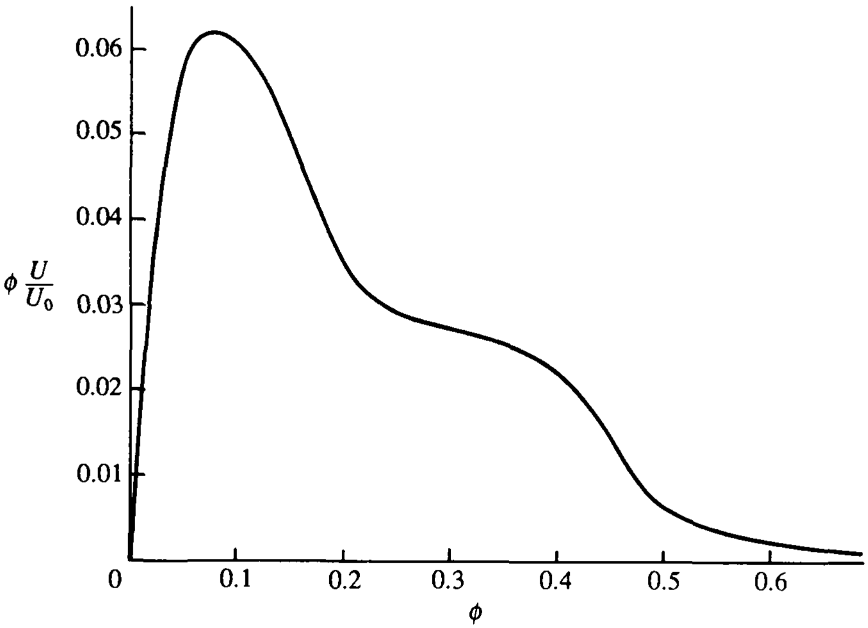
1 Approximate K_2 for hard spheres by evaluating (12.3.1) with the analytical far-field forms for the hydrodynamic functions.

2 Use the results for the adhesive excluded-shell potential (12.3.4) to approximate K_2 for polystyrene latices with $a = 1.0\,\mu\text{m}$ and $\psi_s = 50\,\text{mV}$ for ionic strengths of 10^{-3}–10^{-1} M.

3 Combine the results (12.4.4), (12.4.5), and (12.4.7) into approximate expressions for the settling velocities in a bimodal mixture concentrated in large spheres but dilute in small ones. Identify the nature of the dominate interactions.

4 Formulate the equations governing the transient settling of a bimodal mixture at infinite Peclet number. Then construct solutions via the method of characteristics for $\lambda = 0.1$ and $\gamma = 1.0$, with the initial concentrations $\phi = 0.05$ and $\phi_2 = 0.005$. Assume a sediment volume fraction of 0.55 for the larger spheres, independent of the volume fraction of the smaller spheres.

5 Construct the solution for monodisperse spheres with the flux curve given below and $\phi_m = 0.64$ settling at infinite Peclet number from an initial dispersion with $\phi_0 = 0.15$.



6 Derive the solution for the volume fraction within an equilibrium sediment, i.e. $\phi(x, \infty)$, from (12.5.7). Plot the profiles resulting from $Pe = 10$, 10^2, and 10^3 for hard spheres initially at $\phi = 0.1$.